

Reimagining the CGI Pipeline with Diffusion Models: A Renderless Workflow for 3D Product Visualization

Athirah Zaini

Faculty of Engineering and Design
University of Auckland
Auckland, New Zealand
amoh497@aucklanduni.ac.nz

Allan Fowler

Faculty of Engineering and Design
University of Auckland
Auckland, New Zealand
allan.fowler@auckland.ac.nz

Auzaei Saffiey

Content Creation
Fisher & Paykel Appliances Limited
Auckland, New Zealand
auzaei.saffiey@fisherpaykel.com

Xue Bai

Faculty of Engineering and Design
University of Auckland
Auckland, New Zealand
xbai405@aucklanduni.ac.nz



Figure 1: Input Images: Stock images of product and room under varied lighting [Africa nd; Khalid nd]. **Output Image:** Product seamlessly composited into the scene. **Output Video:** Generated video showing perspective and lighting shifts.

Abstract

This paper presents a workflow that reimagines the traditional CGI pipeline for product visualization by leveraging FLUX. 1 Kontext and WAN 2.2 diffusion models. Our approach streamlines the creation of high-quality product videos, eliminating time-intensive 3D workflow while seamlessly merging visual elements and harmonizing lighting and aesthetics. This enables rapid look development and iterative pre-visualization, offering faster turnaround and greater creative flexibility than conventional render-based workflows.

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1 Introduction

Product visuals are often the first point of contact in e-commerce and online marketing, making beauty shots vital for attracting and persuading consumers. Traditionally, creating these shots requires

a complex and resource-intensive CGI pipeline, encompassing 3D modeling, UV unwrapping, texturing, HDRI lighting rigs, camera tracking, light matching, 3D rendering, and compositing. This process becomes a bottleneck under tight deadlines or when rapid iterations are required.

Today, 3D Artists increasingly use AI to accelerate their workflows. However, many models, particularly text-to-video, struggle to maintain product accuracy, spatial coherence, and fine detail across frames. To address this, we propose a modular "Artist-guided workflow" that bridges this gap, enabling efficiency and control in content generation.

We use FLUX. 1 Kontext, a unified generative model that integrates image creation and editing from text and image inputs [Labs et al. 2025], supporting local and global edits while maintaining object consistency across iterations. We also use WAN 2.2, a diffusion-transformer video foundation model that achieves high-quality generation through its variational autoencoder (VAE), scalable pre-training techniques, extensive data curation, and automated evaluation metrics [Wan et al. 2025].

2 A Two-Stage Generative Pipeline

We demonstrate a modular two-stage generative workflow (Figure 2) designed to composite and transform a static product image into a dynamic video. This modular design is a strategic choice that isolates the distinct challenges of spatial synthesis (image compositing) from temporal synthesis (video sequence). By separating these concerns, we can deploy highly specialized models at each stage,

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allowing for targeted optimization and quality control difficult to achieve in a single, end-to-end model. Stage 1 employs Black Forest Labs' FLUX. 1 Kontext [Labs et al. 2025] with a corrective LoRA stack, while Stage 2 uses Alibaba's WAN 2.2 diffusion model [Wan et al. 2025] for video synthesis.

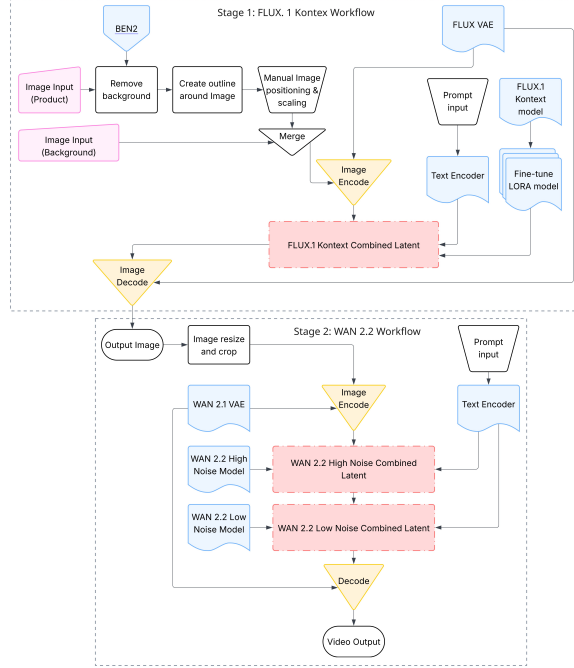


Figure 2: Visual diagram of Stage 1 and 2.

2.1 Stage 1: Image Compositing with FLUX. 1 Kontext and a Corrective LoRA Stack

The objective of the Stage 1 workflow is to embed an isolated product into a new, photorealistic scene while actively mitigating common generative artifacts. Our pipeline begins with two input images: A studio-lit black washer and a lifestyle laundry room image, each under different lighting conditions.

We first use the BEN2 model [Meyer and Spruyt 2025] to isolate the black washer from its original background. It is then manually positioned and scaled to replace the original appliance in the target scene (laundry room). Subsequently, the FLUX VAE encodes this composition into a latent representation, which becomes a key input for this two-stage workflow. This latent, along with a descriptive text prompt, is fed into the FLUX. 1 Kontext model. While we selected FLUX. 1 for its robust in-context editing, its generalist nature can produce artifacts. To address this, we integrate a dual-LoRA stack for targeted guidance. We apply the Put-it-here-Kontext-v0.2 LoRA for task specialization by embedding its required trigger word within the prompt. It directs the FLUX.1 base model to focus on blending the product and ensuring plausible lighting and shadows. Next, we apply the UNFLUX LoRA as a corrective measure to reduce the visual distortions and artifacts common to FLUX models.

The effectiveness of this process is demonstrated in the output image (Figure 1). The new product is seamlessly integrated, inheriting the scene's soft ray light and casting a physically consistent contact and ambient shadow, resulting in a high-quality, artifact-free image ready for video synthesis.

2.2 Stage 2: Video Synthesis with WAN 2.2

Stage 2 takes the static, composed image and introduces motion with a perspective shift. The image is first encoded into a latent representation by the WAN 2.1 VAE. Video synthesis is then performed by the WAN 2.2 model, guided by a text prompt dictating the desired motion. We selected WAN 2.2 for its Mixture-of-Experts (MoE) architecture, which enables efficient, high-quality generation. The model employs a coarse-to-fine refinement strategy, where a high-noise model establishes foundational motion vectors, which a low-noise model then refines by adding details and ensuring frame-to-frame consistency [Wan et al. 2025]. The final refined latent is then synthesized by a decoder into the output video. The resulting five-second video shows smooth temporal consistency according to the motion prompt and provides a view of the washer's interior, bypassing the need for traditional 3D modeling workflows.

3 Conclusion

This paper presents a generative, artist-guided workflow as a viable alternative to the traditional, render-intensive CGI pipeline for 3D product visualization. Using a modular, diffusion-based approach, we bypass the time-consuming stages of 3D modeling and rendering. The workflow is demonstrated through a case study compositing a black washer into a lifestyle scene. An example that typifies challenges in 3D appliance visualization, such as lighting harmonization, shadow realism, and object integration, which provides a suitable benchmark for evaluating our renderless workflow.

Performance was evaluated on a consumer-grade RTX 3080 Ti mobile GPU, where VRAM constraints established 480p as the stable testing resolution. 30 video generation trials were conducted and evaluated for visual fidelity, artifact presence, and spatial motion consistency. 21 trials were successful (70% success rate), with an average generation time of 18.4 minutes per video. Observed failure included human hand intrusion (n=1), washer drum noise (n=3), dark shadow artifacts (n=3), background distortion (n=1), and incorrect camera motion (n=1). In contrast, traditional CGI workflows typically require 2–5 days of a senior artist's time to achieve comparable fidelity, demonstrating a substantial reduction in production time and manual effort. This demonstrates that a renderless, artist-guided workflow can accelerate the creative process, enabling rapid iteration on concepts, previsualizations, and final deliverables, while maintaining art-directable control and professional outputs.

While our approach enables unprecedented speed, it remains constrained by current hardware, limiting resolution and video duration. However, ongoing advancements in generative models are expected to improve both performance and quality. The modular design of our workflow enables seamless integration of such improvements. Future work will extend the approach across diverse product categories and environments, including higher-resolution testing to assess robustness and versatility.

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